

JULIUS AUSTIN S. SANTOS

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EDUCATION

University of Santo Tomas

Expected Graduation: 2027

Bachelor of Science in Information Technology (BSIT) - College of Information and Computing Sciences (CICS)

Manila, Philippines

- Cumulative GWA: 1.269
- Academic Standing: Consistent Dean's Lister (Top 1 – 2nd Year; Top 4 & 8 – 1st Year; 3rd Year Dean's Lister)

Pateros Catholic School

2021 – 2023

Senior High School — Science, Technology, Engineering, and Mathematics (STEM)

Pateros, Metro Manila

- Graduated with Honors (Academic Honors in Grade 11 & Grade 12)

Pateros Catholic School

2017 – 2021

Junior High School

Pateros, Metro Manila

- Graduated with Honors (Consistent Academic Honors from Grades 7 to 10)

TECHNICAL SKILLS

- **Operating Systems & Administration:** Linux (Ubuntu, AlmaLinux), Windows, Windows Server, Active Directory
- **Hardware, IoT & Embedded Systems:** PC Building & Hardware Diagnostics, Component-Level Troubleshooting, Raspberry Pi 5, ESP32, Arduino Uno/Mega, mmWave Radar integration
- **Cloud, Hosting & Networking:** Hostinger Deployment & Web Hosting, Cisco Packet Tracer, Aviatrix Multicloud, Oracle Cloud Infrastructure (OCI)
- **Programming & Databases:** Python, Java, Kotlin, PHP, JavaScript, C++, C# SQL, MySQL, Firebase Firestore
- **Web & Mobile Development:** Laravel, Android Studio, React.js, Svelte, Tauri, FastAPI, Bootstrap, HTML5, CSS3
- **AI Integration & Developer Tools:** Gemini AI API Integration, RESTful APIs, Git/GitHub, Linux CLI, Autodesk Fusion, Figma, Tinkercad, Unity Engine

KEY TECHNICAL PROJECTS

Internet-Independent Multimodal Life Detection & Localization SAR Rover

2026

Hardware Lead, Backend Developer & CAD Enclosure Designer | Edge AI & Robotics

- Designed and integrated an edge-computing SAR payload on a mobile rover using a Raspberry Pi 5, Seede 60 GHz FMCW mmWave radar (MR60BHA2), XIAO ESP32, Pi Camera, and NEO-M8N GPS module.
- Engineered custom 3D-printed modular enclosures with mechanical mounting tolerances for dual-band wireless HDMI TX/RX, sensor arrays, and battery power units using CAD modeling.
- Developed an internet-independent, full-stack local web application using FastAPI and Svelte to display live video feeds, real-time sensor metrics, and location tracking.
- Designed the frontend interface featuring dynamic telemetry graphs (Chart.js), real-time vital-sign displays, GPS logging, and WebSocket-driven status alerts.
- Implemented multimodal sensor fusion logic combining SSD MobileNet V3 object detection with radar bandpass filters (0.1–0.66 Hz respiration, 0.5–3.33 Hz heartbeat) to trigger real-time survivor warning alerts.

Integrated Information Management System (Reality-Based TC Inc.)

2025 – 2026

Full-Stack Developer & UI/UX Designer | Laravel, PHP, MySQL, Bootstrap, Hostinger

- Designed UI/UX mockups in Figma, planned the system workflow, and coded the full-stack web app using Laravel, PHP, and MySQL.
- Built a secure patient record system with role-based access control (RBAC) to protect confidential data.
- Created a medicine tracking system to monitor pharmaceutical stock and keep clear logs of medications given to each patient.
- Deployed the live web app and database on Hostinger.

ShelfCare: Smart Community Kiosk & Mobile Donation System

2025

Mobile Developer & Hardware Integrator | Android Studio, Gemini AI, Firebase, Arduino

- Developed an Android mobile application integrated with the Gemini AI API for computer vision-based item recognition of donated goods in smart drop-off bins.
- Integrated the Google Maps API to provide real-time location tracking and navigation to nearby community donation kiosks.
- Built an interactive hardware prototype using Arduino, push buttons, sensors, and servo motors to control automated donation bin access and status alerts.
- Configured Firebase Authentication and Firestore to manage donor user profiles, inventory updates, and donation activity logs.

LeSanShoes: Full-Stack E-Commerce Web Application

2025

Full-Stack Backend Developer | PHP, MySQL, JavaScript, Bootstrap, CSS3

- Developed a full-stack shoe store e-commerce application featuring product browsing, dynamic catalog management, and order placement.
- Implemented CRUD operations and administrative backend workflows to update shoe inventory, pricing, and customer orders.
- Structured the relational MySQL database and handled front-to-back integration for seamless cart management and checkout simulation.

Disaster Run: 2D Action Platformer Game (SHS STEM Capstone)

2023

Game Developer & Animator | Unity Engine, C#

- Programmed player movement mechanics, custom physics, and collision logic in C# for a 2D action platformer capstone game.
- Designed and implemented 2D sprite animations, state transitions, and responsive user interfaces within the Unity Engine.

Additional Web & Software Engineering Portfolio Projects

2024 – 2025

Developer | Full-Stack & Systems

- Tiyarjiz Rug E-Commerce Website: Built responsive full-stack commercial web platforms featuring custom catalog filters and checkout flows.
- Purrfect & JolliDine Portals: Developed modern, accessible front-end interfaces utilizing responsive Bootstrap components and vanilla JavaScript.
- DSA & OS System Projects: Implemented core memory management algorithms, process scheduling simulations, and data structure benchmarks in C++/Java.

LEADERSHIP & CLASS OFFICERS

UST College of Information and Computing Sciences (CICS)

2023 – Present

Elected Class Officer

Manila, Philippines

- Class Vice President (4ITC 1st Sem, 3ITC 1st & 2nd Sem): Assisted the class president in disseminating announcements, stepped in to lead during the president's absence, and relayed student academic inquiries and block concerns to course professors.

- Class Auditor (IITG 2nd Sem) & Class Treasurer (IITI 1st Sem): Collected class contributions for PathFIT uniforms and academic fees, kept balanced financial ledgers, and verified zero discrepancies in classroom fund disbursements.

Society of Information Technology Enthusiasts (SITE - UST)

2023 – Present

Executive Coordinator / Associate Team Head / Committee Staff

Manila, Philippines

- Executive Coordinator for Secretariat (3rd Year): Handled event documentation and paperwork across organization committees, coordinating directly with event point persons for smooth logistics.
- Associate Team Head for Academic and Research (2nd Year): Monitored the creation and progress of IT department reviewers, helped manage departmental events like the IT Research Colloquium, and surveyed student feedback on challenging subject topics.
- Committee Staff for Academic and Research (1st Year): Drafted comprehensive academic reviewers and study materials on schedule to support IT block mates and lower-batch students.

Student Organizations & Affiliations

2023 – Present

Active Member / Committee Staff

Manila, Philippines

- UST AWS Student Community: Member engaging in cloud computing architectures and serverless workshops (2025 – Present).
- UST UNICEF Volunteers (UST UVU): Member supporting community outreach and advocacy initiatives (2024 – 2025).
- Cisco Networking Academy Guild (CNAG): Environmental Committee Staff advocating green computing practices (2023 – 2024).

HONORS & ACADEMIC AWARDS

Dean's List Academic Honors

2023 – 2026

UST College of Information and Computing Sciences

- Rank 1 (Top 1) – 2nd Year, 1st Semester & 2nd Semester
- Rank 4 (Top 4) – 1st Year, 1st Semester | Rank 8 (Top 8) – 1st Year, 2nd Semester
- Dean's Lister – 3rd Year, 1st Semester & 2nd Semester

COMPETITIONS & TECHNICAL ACTIVITIES

Hackathons & Technical Competitions

2024 – 2025

Participant / Finalist

- Finalist – EDUtech Asia Planet Protectors: 2025 Sustainability Challenge (Singapore)
- Participant – UST Hack-a-Thon: Innovating for Health, Environment, and Food (Nov 2024 – Jan 2025)
- Participant – UMAK IT Skills Challenge E-Robotics Challenge (2025)
- Participant – IT Code Clash Programming Competition (May 2024)

Technical Event Production & Volunteering

2023 – 2025

Technical Staff & Volunteer

- Technical Staff & Production Assistant – IT Research Colloquium (Apr 2024 & Apr 2025)
- Technical Staff – IT Right on Track & IT Kickoff (Apr & May 2024)
- Student Volunteer – CICS CARE Outreach Program (Nov 2023)

CERTIFICATIONS & PROFESSIONAL BADGES

- CompTIA IT Fundamentals+ (ITF+) Certification (CompTIA, May 2024)
- Aviatrix Certified Multicloud Network Associate (Aviatrix, Active thru July 2028)

- Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate (Oracle, Aug 2025)
- Databases and SQL for Data Science with Python (IBM & Coursera, Dec 2024)
- AI Fundamentals (IBM SkillsBuild, Dec 2024)
- Computer Hardware Basics & Introduction to IoT (Cisco Networking Academy, 2024 – 2025)
- Linux Unhatched (Cisco / NDG, Jan 2025) & Developing Websites with Bootstrap (Dec 2024)

SEMINARS & LEADERSHIP TRAINING

- Cybersecurity Awareness Learning Session – University of Santo Tomas
- Peer Facilitator Training – UST Guidance and Counseling Department
- CICS Week Technical Learning Series & Seminars – UST CICS (2024 & 2025)